

Auditable Instruments Dominate Language Models on Calibrated Geopolitical Assessment

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Abstract

If an LLM can draft a geopolitical brief, does it replace the scoring stack behind serious OSINT? We show that it does not. On three frozen event packs, we evaluate **Claude Haiku 4.5** and **GPT-4o-mini** (temperature 0, schema-locked JSON) against ARGUS open instruments—absolute country risk, rate anomalies, offline Analysis of Competing Hypotheses (ACH), and evidence-balance confidence caps—using identical inputs. **Where assessment requires calibration, instruments dominate.** Absolute risk-band agreement collapses to 0.11 for both models. ACH cell agreement remains middling (Claude 0.40, GPT 0.53). GPT violates confidence caps (0.33 overconfident); Claude is more restrained but still imperfect. Models look strong only on *easy* axes: anomaly labels (1.00) and risk *rank* (GPT $\rho = 1$). Ordinal fluency is not calibrated assessment. Auditable instruments should own bands, ACH cells, and confidence; LLMs may assist triage and prose. Replication materials accompany this submission.

Keywords: OSINT evaluation, geopolitical risk, ACH, LLM evaluation, calibration, auditable scoring

1 Introduction

Large language models are increasingly treated as geopolitical analysts: ranking theaters, filling ACH matrices, and stating confidence. Fluency makes that pitch easy. Tradecraft makes it false unless the model matches *inspectable* methods on the axes that decide whether a brief is usable.

This paper tests a dominance claim:

On fixed OSINT packs, ARGUS instruments dominate schema-locked LLMs on calibrated assessment tasks (absolute risk bands, ACH cells, confidence caps). LLMs match instruments only when the task reduces to following a rate rule or recovering rank order.

We do **not** claim that instruments equal world ground truth. We claim something stronger for *systems*: instruments are **reproducible authorities**—same inputs, same numbers, no vibe. That is the standard LLMs fail here.

Table 1: ARGUS instruments used as the reference authority.

Task	Instrument	Authority output
Risk ranking	<code>projectRisk</code>	Score 0–100 + CRITICAL / HIGH / MEDIUM / LOW
Anomaly call	<code>rateZScore</code> + thresholds	Surge + none / mild / strong
ACH matrix	<code>scoreACHOffline</code>	supports / neutral / contradicts
Brief confidence	<code>assessEvidenceBalance</code>	<code>confidenceCap</code>

Thresholds were locked *ex ante*. Instruments are the reference; models are the challengers.
Anomaly rule: $z \geq 3$ strong; $1.5 \leq z < 3$ mild; else none.

Contributions.

1. A locked, open replication package (datasets, prompts, schemas, runners, scored outputs).
2. Head-to-head results: Claude Haiku 4.5 and GPT-4o-mini versus ARGUS instruments on four tasks $\times$ three packs.
3. A task-wise dominance map: instruments win calibration; models win only soft ordinal / numeric-follow tasks.

2 Related work

Event data and risk scoring. Structured streams such as ACLED [3] and GDELT [2] underpin political-risk practice. ARGUS implements absolute country risk from severity, fatalities, and related composites (`lib/riskScoring.ts`, `lib/projectRisk.ts`).

ACH. Heuer’s Analysis of Competing Hypotheses [1] remains the falsification backbone of analytic tradecraft. We use ARGUS offline ACH (`lib/offlineIntel.ts`) as an auditable matrix—not human gold, but a fixed rule baseline LLMs must beat to claim ACH competence.

LLM evaluation and confidence. Prior work stresses schema adherence, calibration, and overconfidence. We operationalize those concerns on geopolitical instruments. Thin corpora must clamp confidence; ARGUS evidence-balance (`lib/evidenceBalance.ts`) emits an explicit HIGH / MODERATE / LOW cap.

3 Instruments (the authority set)

4 Experimental design

4.1 Datasets

Three frozen synthetic packs (license-clean; no live-API drift) are summarized in Table 2. Full JSON packs are in the supplement data directory.

4.2 Challengers

Shared caps: at most 12 events per prompt; completion limit 800 tokens (2000 for ACH).

Table 2: Evaluation packs.

Pack	Character	Events	Instrument anomaly
conflict-rich	Kinetic-heavy frontier	8	$z=6.73$ (strong)
maritime-mixed	Maritime + economic mix	4	$z=1.73$ (mild)
thin	Near-empty political corpus	2	$z=0$ (none)

Table 3: Model conditions (temperature 0; shared prompts and schemas).

Arm	Model ID	Effort
Claude	claude-haiku-4-5-20251001	low
OpenAI	gpt-4o-mini	low

4.3 Metrics (how dominance is scored)

5 Results

5.1 Instrument baselines

Deterministic instrument outputs (\$0 API) appear in Table 5.

5.2 Dominance table (primary claim)

Table 6 is the main empirical claim of the paper.

5.3 What failure looks like

GPT assigned Ukraine **CRITICAL** / **85** on the maritime pack while the instrument scored **LOW** / **14**—same events, different authority. ACH cell agreement of 0.40–0.53 means roughly half the falsification matrix disagrees with the offline method. GPT stated **HIGH** confidence on the conflict-rich pack against a **MODERATE** evidence cap.

6 Discussion

1. **Bands and ACH define the product.** Briefs live or die on absolute levels and hypothesis cells; instruments own both here.
2. **Anomaly “wins” for models are cheap.** Matching z -thresholds shows schema following, not analytic superiority.
3. **Rank $\neq$ risk.** High Spearman with weak band agreement is the strongest warning result.
4. **Confidence is the tell.** Even when Claude avoids overclaim, removing the instrument cap is unjustified.

Systems implication: ARGUS-style stacks should compute scores first and let models narrate second—not the reverse.

Table 4: Win conditions used in the dominance map.

Task	Model “wins” if...	Instrument dominates if...
Risk bands	High band agreement	Band agreement near zero / systematic inflation
Risk rank	High Spearman ρ	Secondary only (ordinal)
Anomaly	Surge + severity match	Model fails the rate rule
ACH	High cell agreement	Cell agreement far below usable quality
Confidence	Cap match; no overclaim	Overclaim or miss the cap

Table 5: Instrument baselines by pack.

Pack	Top risk	Evidence cap	Anomaly
conflict-rich	Ukraine 46 / HIGH	MODERATE / 52	strong
maritime-mixed	Ukraine 14 / LOW	MODERATE / 63	mild
thin	Unknownland 1 / LOW	MODERATE / 63	none

7 Limitations

Pilot scale ($n=3$ packs); cost-tier models (Haiku / 4o-mini); offline ACH is not human gold; synthetic English short-form events only. Larger models may narrow gaps; that would be an empirical update, not a reason to skip instruments today.

8 Conclusion

Auditable instruments dominate language models on calibrated geopolitical assessment under this protocol. Claude and GPT match rate-based anomalies and can recover risk order; they do **not** match absolute bands or ACH cells at usable levels, and GPT overclaims confidence when the corpus looks rich. The correct architecture is instrument-authoritative scoring with optional LLM assistance—not LLM-authoritative scoring with optional formulas.

Acknowledgments

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References

- [1] Richards J. Heuer. *Psychology of Intelligence Analysis*. CIA Center for the Study of Intelligence, 1999.
- [2] Kalev Leetaru and Philip A. Schrodt. GDELT: Global data on events, location and tone. Technical report, International Studies Association Annual Convention, 2013.
- [3] Clionadh Raleigh, Andrew Linke, Håvard Hegre, and Joakim Karlsen. Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5):651–660, 2010.

Table 6: Task-wise dominance map ($n=3$ packs per model).

Axis	Claude	GPT	Winner
Absolute risk bands (agree)	0.11	0.11	Instrument
Risk rank ρ	0.75	1.00	Model (ordinal only)
Anomaly surge / severity	1.00 / 1.00	1.00 / 1.00	Tie (rate-rule follow)
ACH cell agreement	0.40	0.53	Instrument
Confidence cap match	0.67	0.33	Instrument
Overconfidence rate	0.00	0.33	GPT fails; Claude restrained

Models do not overturn instruments on any calibrated axis.

High Spearman with $\sim 11\%$ band agreement is *false competence*: same order, wrong magnitude.